NPR College of Engineering and Technology
(AN AUTONOMOUS INSTITUTION)
NBA Accredited (B.E. - CSE, ECE, EEE & Mechanical Engg.) | Accredited by NAAC with 'A' Grade
Recognized by UGC under 2 (f) | ISO 9001:2015 Certified | Approved by All India Council for Technical Education, New Delhi
Affiliated to Anna University, Chennai
Natham, Dindigul (Dist.) - 624 401, Tamil Nadu.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT WORK

FIRST REVIEW

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PROJECT TITLE : Fake Review Detection Using Hybrid Transformer and Behavioral Analysis

BATCH NUMBER:

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PROJECT GUIDE:

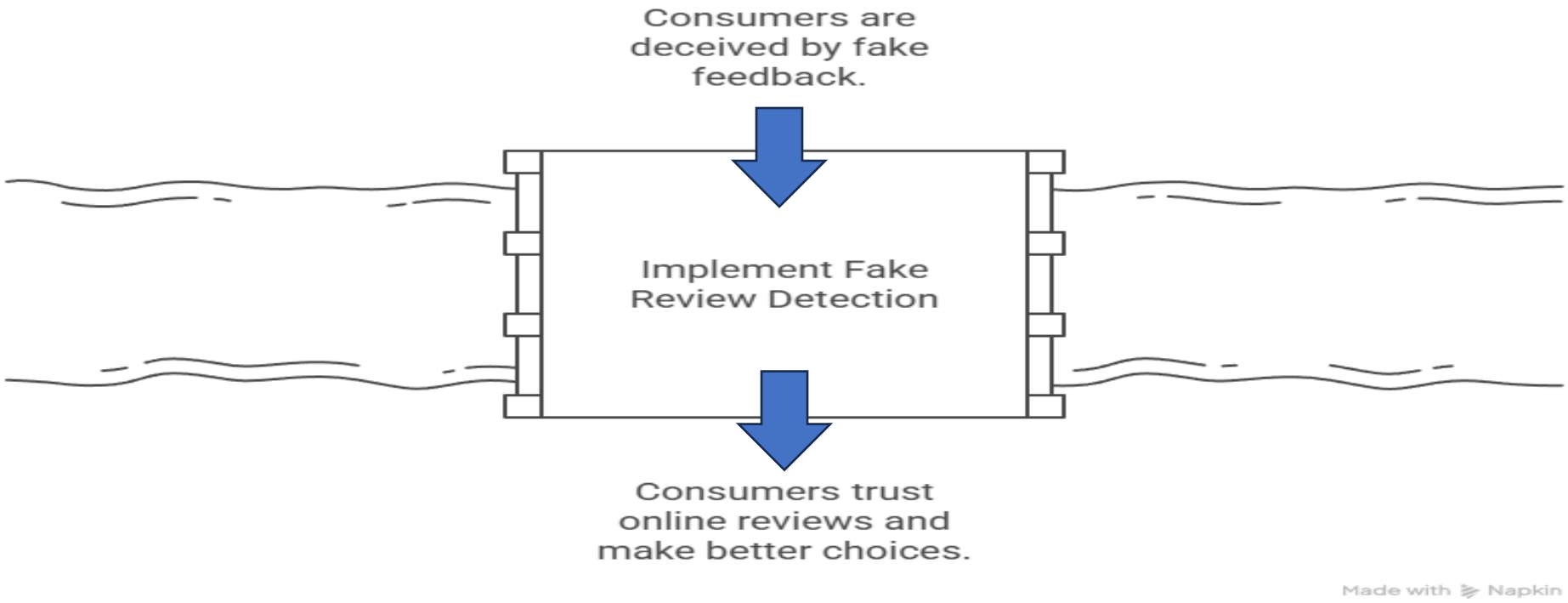
ABSTRACT:

- Online reviews strongly influence customer decisions in e-commerce platforms. However, the rapid increase of fake and AI-generated reviews has reduced consumer trust.
- This project proposes an advanced Fake Review Detection System that integrates:
  - Transformer-based language models (BERT)
  - Linguistic feature extraction
  - User behavioral analysis
  - Explainable AI techniques
- The system improves detection accuracy, reduces false negatives, and provides interpretable results. The proposed hybrid approach enhances reliability, scalability, and real-world applicability.

INTRODUCTION:

- Online reviews play a crucial role in influencing consumer buying decisions, with over 80% of consumers relying on them when making purchases. However, the rise of fake reviews has become a significant concern, as they can manipulate product rankings, damage brand reputation, and erode customer trust.
- These fake reviews can be generated by various sources, including paid individuals, automated bots, and AI-based text generators.
- Current detection systems primarily focus on textual analysis, which highlights the need for more sophisticated hybrid detection models that incorporate behavioral pattern analysis and AI-generated content detection.
- This document outlines the limitations of existing systems and introduces a proposed solution that addresses these challenges.

Fake reviews mislead consumers; a detection system restores trust.



EXISTING SYSTEM:

Current fake review detection systems typically utilize techniques such as TF-IDF, n-grams, sentiment analysis, and traditional machine learning models like Logistic Regression, Support Vector Machines (SVM), and Random Forest. Additionally, they rely on basic behavioral features to identify fraudulent reviews.

Limitations:

- Limited contextual understanding of reviews.
- Poor detection capabilities for AI-generated content.
- Lack of explainability in results.
- Performance degradation with imbalanced datasets.
- Weakness in real-time deployment capabilities.

PROPOSED SYSTEM:

Our proposed system integrates several advanced components to enhance fake review detection:

- **Transformer-Based Text Analysis**: Utilizing BERT for contextual embeddings allows for deep semantic understanding of the text.
- **Behavioral Feature Extraction**: This includes analyzing review frequency, rating patterns, account age, and detecting suspicious activities.
- **Feature Fusion Layer**: This layer combines both textual and behavioral features to improve classification accuracy.
- **Explainable AI Module**: Implementing SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) for interpretation and word-level importance visualization.
- **Performance Evaluation Metrics**: The system will be evaluated based on accuracy, precision, recall, F1 score, and AUC (Area Under the Curve).

Fake Review Detection System Process

Transformer-Based Text Analysis

BERT is used for deep semantic understanding of text.

Behavioral Feature Extraction

Review frequency, rating patterns, and account age are analyzed.

Feature Fusion Layer

Textual and behavioral features are combined for accuracy.

Explainable AI Module

SHAP and LIME are used for interpretation and visualization.

Performance Evaluation Metrics

Accuracy, precision, recall, F1 score, and AUC are used for evaluation.

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LITERATURE SURVEY:

A review of relevant literature reveals several significant contributions to the field of fake review detection:

- **Ott et al. (2011)**: This study utilized linguistic features to effectively detect deceptive reviews.
- **Mukherjee et al. (2013)**: Introduced a behavioral-based spam detection method using Yelp data, highlighting the importance of user behavior in identifying fake reviews.
- **Lim et al. (2010)**: Focused on analyzing rating behavior patterns as a means of spam detection.
- **Zhang et al. (2016)**: Compared verbal and non-verbal features in the context of fake review detection, providing insights into the multifaceted nature of deceptive reviews.
- **Recent Transformer-based Approaches (2020–2023)**: These studies demonstrate that contextual embeddings significantly enhance detection accuracy, underscoring the potential of hybrid deep learning approaches.

Conclusion from Literature:

- Most existing works tend to use either textual or behavioral features in isolation. The evolution of hybrid deep learning approaches presents a promising avenue for future research and application in fake review detection.

MODULES DESCRIPTION:

The proposed project can be divided into several key modules:

- **Module 1: Data Collection Module**
  - Collect datasets from platforms like Yelp and Amazon.
  - Label reviews as genuine or fake.
- **Module 2: Data Preprocessing Module**
  - Conduct text cleaning, stopword removal, and tokenization.
  - Address imbalanced data using techniques like SMOTE (Synthetic Minority Over-sampling Technique).
- **Module 3: Feature Extraction Module**
  - Extract text features using BERT.
  - Engineer behavioral features and normalize them.

- **Module 4: Model Training Module**
  - Train various models, including Logistic Regression (as a baseline), Random Forest, and a BERT-based classifier.
  - Perform hyperparameter tuning for optimal performance.
- **Module 5: Evaluation Module**
  - Utilize confusion matrices to assess model performance.
  - Calculate accuracy, F1 score, and generate ROC-AUC curves.
- **Module 6: Explainability & Deployment Module**
  - Conduct SHAP/LIME analysis for model interpretability.
  - Integrate a REST API for deployment and create a web-based prediction interface.

Modules Description for Fake Review Detection System

Data Collection Module

Dataset Acquisition

Collect and label reviews from Yelp and Amazon

Data Preprocessing Module

Text Cleaning & Balancing

Clean text, remove stopwords, tokenize, and balance data with SMOTE

Feature Extraction Module

BERT & Behavioral Features

Extract text features using BERT and engineer/normalize behavioral features

Model Training Module

Classifier Training

Train Logistic Regression, Random Forest, and BERT-based classifiers with hyperparameter tuning

Evaluation Module

Performance Assessment

Evaluate models using confusion matrices, accuracy, F1 score, and ROC-AUC curves

Explainability & Deployment Module

API

Interpretability & API

Conduct SHAP/LIME analysis, integrate REST API, and create web interface

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THANK YOU